ENG1013 Traffic System Assignment

Concise overview extracted from the Traffic System Project Specification v11.1.1 2026.

Mission Statement

Design, build and demonstrate a simplified smart traffic-control safety system for the Blackwall Tunnel southern approach. The system must reduce over-height vehicle collision risk by detecting vehicle height, warning drivers, controlling traffic and pedestrian signals, providing an over-height exit route, and failing safely during tunnel-height subsystem power loss.

The model focuses on the A102 freeway approach, Tunnel Ave, an over-height exit, the tunnel entrance, and a hardware-backed failure alert system. The default vehicle height limit is 4.0 m, with user configuration required in relevant subsystems.

Build Scope

Subsystem	Purpose	Key hardware
1. Approach Height Detection	Primary over-height detection before the exit. Warns, records, and controls TL1/TL2.	US1, US2, TL1, TL2, WL1, PA1
2. Tunnel Ave Control	Controls Tunnel Ave junction and pedestrian crossing, including day/night timing.	TL4, TL5, PL1, PL2, PB1, PB2, DS2
3. Over-height Exit	Lets an over-height vehicle leave via the exit and supports night floodlighting.	US5, TL6, DS1, FL1, FL2
4. Tunnel Height Detection	Last-chance detection at the tunnel entrance with verification and closure warnings.	US3, US4, TL3, WL2
5. Failure Alert	Hardware-only fail-safe that reacts to tunnel-height subsystem power loss.	Battery, OS1, PA2, red-light overrides

Core Constraints

Hardware is limited to the components supplied across the team kits, plus single-core wire and extra breadboards. Maximum integration marks require one Arduino and a single code launch point. If integration features are not targeted, multiple Arduinos may be used, but only one can be active at a time.

Software must use Python 3.10.x. Supported packages are Pymata4, Time, Math, Matplotlib, and Random. NumPy and CSV are permitted but unsupported by demonstrators. Other packages, classes, and assert statements are not permitted for marking. Generative AI use is allowed only with a PDF declaration, except Milestone 3 reflection work where AI use is prohibited.

Milestones and Deliverables

Milestone	Due / event	Main deliverables	Points
M1: Design and plan	End of Week 5, Friday 11:55pm	System Design Document, AI usage declaration, hardware interaction block diagram, subsystem flowcharts, timeline, communications plan, conflict resolution plan.	80
M2: Build and demonstrate	End of Week 10 submission, Week 11 demonstration	Code and circuits for required features, selected general/integration features, circuit diagrams, physical prototype, backup video URLs, single ZIP submission.	180
M3: Reflection and viva	Reflection by end of Week 11, viva in Week 12	Individual reflection quiz, meeting minutes, in-person viva confirming understanding of software and hardware.	Required

Marking Notes

The project is graded out of 260 points and contributes to 26% of the ENG1013 total mark after scaling. Individual scaling uses the third ITP Metrics peer assessment score and viva performance at 50% weighting each. A missed viva means no project marks.

Demonstrated functionality earns full available feature points. A submitted backup video can support 66% if the physical demo fails. If the team can only explain how a failed feature should work, the available mark drops to 33%.

Feature Colour Coding

Required features are marked in blue in the specification, general features are unhighlighted, and integration features are marked in yellow. The running system must continue until KeyboardInterrupt, then turn off pins and shut down the Arduino cleanly.

Milestone 1 Checklist

Submit two PDFs: TeamNNN-M1_AIDeclaration.pdf and TeamNNN-M1_SDD.pdf. The design document must keep the prescribed order: system interaction block diagram, high-level flowcharts, proposed timeline, communications plan, and conflict resolution plan.

The 80 points are split as: system interaction block diagram 20, high-level flowcharts 30, proposed timeline 10, communications plan 10, and conflict resolution plan 10. Common deductions include incorrect format or filename, poor diagram quality, missing components, and plans that are incomplete or unclear.

Milestone 2 Checklist

Submit a single TeamNNN-M2.zip containing a TeamNNN folder with the AI declaration, a single circuit_diagrams.pdf, Python files with no spaces in filenames, and backup_video_url.txt. Circuit diagrams must be supplied for any circuit demonstrated. The physical circuit must be ready for Week 11 demonstration.

M2 allocates 80 points to required subsystem features, up to 80 points to selected general/integration features across subsystem caps, and 20 points to full-system integration on one Arduino through a single code entry point.

Immediate Portfolio Interpretation

For an engineering portfolio, this project currently demonstrates system decomposition, requirement extraction, hardware/software planning, team coordination, and traceability from official requirements to Milestone 1 design artifacts. The build, circuit implementation, and demonstration evidence should be added as the project progresses.

Original Specification Screenshot: System Model

4 System Overview

This project is based on the Blackwall Tunnel (Southern Approach) which exists in the UK. We have simplified the actual system in reality to comprise of the A102, the freeway leading up to the tunnel entrance, and Tunnel Ave, the arterial road running parallel to A102 near the tunnel entrance. The main emphasis of the system is on traffic control and to prevent or minimize overheight vehicle collisions at the tunnel entrance, which only can accommodate up to 4.0m high vehicles. The system provides a overheight detection system on approach, a final overheight vehicle exit and a final overheight vehicle detection system just prior to the tunnel entrance. There are also systems in place to take control of the traffic should power fail in the area, which is run off backup batteries. You can find the feature and subsystem specifications under System Features which describes the system behaviour modelled below. US1/US2 should be capable of reading the actual vehicle height, whereas US3/US4/US4 are only required to detect an over-height vehicle passing.

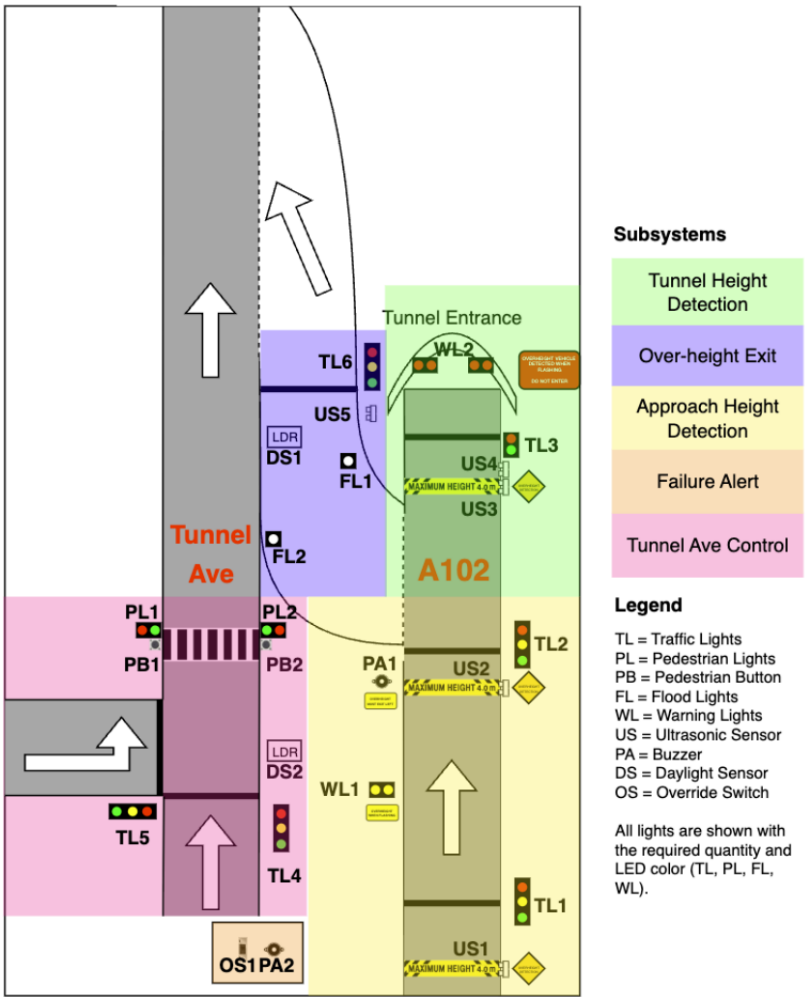

Figure 1: Model of Traffic Control Safety System for the tunnel.

Screenshot from the original assignment specification: simplified Blackwall Tunnel traffic-control model and subsystem layout.

Original Specification Screenshot: Deliverables

5 Deliverables

The project is split into three milestones, where the first milestone is a design/plan stage; the second milestone consists of your system submission; and the third milestone is a reflection stage.

You will earn **points** over the course of this project, which will translate into marks after the completion of the full project (and completion of the viva interview). These marks will be individually scaled based on a Weighted Scaling Factor using your third ITP Metrics Peer Assessment Score (PAF3) at 50% weighting, and your Viva Interview Score at 50% weighting.

You should read section 5 carefully as it outlines your deliverables for each milestone, including all deductions and submission requirements.

The three milestones are as follows:

1. Delivery of a design document (80 points) which includes
    a. Hardware system interaction block diagram showing all interaction between hardware components as blocks
    b. High level software logical flowcharts showing all interaction between components, one flowchart for each subsystem
    c. Planned timeline of work (gantt chart)
    d. Communications plan
    e. Conflict resolution plan
2. Full system submission, demonstration and viva (oral interview) (180 points)
    a. Circuit diagrams (one circuit per function / per page as a .pdf file) for all components being demonstrated
    b. Software code (.py python files) for all work being marked
    c. Physical circuit (on demonstration day in week 11)
3. Project reflection
    a. Complete the reflection task set
    b. Complete the in-person viva in week 12

Your system **must be demonstrated in person** during the assigned session time **to achieve the maximum points available** (100%). However, suppose some part (or whole) of your prototype is not functional on the day. In this case, you use a **video your team submitted as part of your milestone on Moodle** to be marked instead for **66%** of the points available. If you have not submitted a video, your team may **explain** how your system should work for **33%** of the available points.

Individual bonus points

Knowledge gained from completing the Practical Learning Competency tasks will help you greatly throughout the project. If you complete these **within the first two weeks of the assessment offering**, you will be awarded bonus marks for each, to be applied individually to your final project mark (capped at 26%).

Screenshot from the original assignment specification: milestone structure and deliverables summary.

Original Specification Screenshot: M2 Build Scope

5.2 Milestone 2

Milestone 2 is due at the end of Week 10 on Friday at 11:55pm, with demonstration in Week 11, and viva (Milestone 3) in Week 12.

5.2.1 Milestone Deliverables

Your team must demonstrate a functioning system (180 points) that consists of the following:

- Code and circuits implementing required features (in blue) of following subsystems (80 points):
  - Subsystem 1 (Approach Height Detection Subsystem)
  - Subsystem 2 (Tunnel Ave Control Subsystem)
  - Subsystem 3 (Over-height Exit Subsystem)
  - Subsystem 4 (Tunnel Height Detection Subsystem)
  - Subsystem 5 (Failure Alert Subsystem) (*note, this is a hardware only subsystem*)
- Additional code and circuits implementing general and/or integration features from the five subsystems as selected (80 points). A point cap applies to each subsystem as follows for general and integration features selected for implementation (caps apply based on where the feature is specified):
  - Subsystem 1 (Approach Height Detection Subsystem): 10 points
  - Subsystem 2 (Tunnel Ave Control Subsystem): 20 points
  - Subsystem 3 (Over-height Exit Subsystem): 10 points
  - Subsystem 4 (Tunnel Height Detection Subsystem): 20 points
  - Subsystem 5 (Failure Alert Subsystem): 20 points
- Circuit Diagrams for all implemented features. You will not be allowed to demonstrate circuits for which there are no circuit diagrams submitted.
- Full system integration (of all subsystems) as a single code-base (meaning a single code launch execution point) and operating on a single Arduino (20 points).

Points caps are applied to each subsystem's general and integration features - you can implement more than the cap amount of points but you will at most earn that cap amount from the subsystem.

Deductions will only be applied after the total points earned is calculated.

Note that for full system integration marks (20 points), your code must execute from one single starting point (i.e. running the main.py file will start the entire system) and run off a single Arduino for your project demonstration. Any subsystems that are partially integrated will be assessed as a single unit (marked together).

Your team must demonstrate the built system's functionality to your demonstrator in your Week 11 class.

Screenshot from the original assignment specification: Milestone 2 build, circuit, code, and integration requirements.

Original Specification Screenshot: Feature Rules

7 System Features

This section outlines the functional requirements of the system, divided into five subsystems.

The system should continue to run, uninterrupted, until the user chooses to exit (via KeyboardInterrupt).
You must handle the KeyboardInterrupt and cleanly turn off all pins, then shutdown the Arduino.

For your convenience, the features in each subsystem are color coded as follows:

- Required features are shown in **blue highlights**.
- General features are shown in normal text (no highlight).
- Integration features are shown in **yellow highlights**.

Screenshot from the original assignment specification: feature coding rules and clean shutdown requirement.